AI ACADEMY · LEVEL 6 · GRADE 11 · WEEK 24 · STUDENT LESSON

Release and rollback

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

THE CONNECTED WEEK

Match the versions. Respect every required gate.

Explain compatible recovery and why higher correctness cannot cancel a mandatory failure.

Keep one habit: Manifest and contracts → schema matrix → privacy-gate picture → dependency probes → pure decision model → independent prompt/corpus case.

MAIN LESSON · SECTION 1

Review the whole release and its recovery dependencies

> A better score cannot cancel a mandatory failure or an incompatible reader.

Learning goals

I can explain the **mechanism** and terminology of release and **rollback**.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Retrieve versions and environment evidence

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Deployment environments. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Caching and stale information. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Untrusted **input** boundaries. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Prerequisite and starting response

- Recall Deployment environments. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

> **My starting explanation**

End goal

Explain release and **rollback** using the deciding step in this fictional service.

▼ **Earlier wording, alternatives and picture transcripts**

Your learning target

Remember before reading

This week you will practise:

Explain the **mechanism** and terminology of release and **rollback**

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Retrieve what you already know

This week's end goal

Treat release and rollback as a dependency problem

> A single component or score cannot stand in for the complete reviewed combination.

Begin with a homework app and changed file format

Imagine a new homework app saves answers in a new file format. Installing the old app again does not make it able to read the new files. A release is a connected set of code, configuration, data format, prompt, source collection and evaluation evidence. Before approving a change or recovery, check the combination and every required gate. Keep the original failure evidence while testing a repair.

Define release records and current-data compatibility

Release records bind code, configuration, data schema and evaluation **evidence**. **Rollback** requires **compatibility** with the current stored data.

Complete manifest, inventory and gate contract

Declared release or rule	Complete fictional classroom contract
Release A manifest	Code v1; cfg 1; schema 1; prompt p1; corpus c1; eval-A
Release B manifest	Code v2; cfg 2; schema 2; prompt p2; corpus c2; eval-B
Main reader compatibility	v1 reads only schema 1; v2 reads schemas 1 and 2
Main current inventory	Observed record schemas [1, 2, 2] after v2 writes
Main code-only proposal	Return to v1 without changing current data: block schema 2 incompatibility
Separate picture correctness	A 8/10 = 80%; B 9/10 = 90%; these cases only
Mandatory picture privacy gate	B fails; release blocked regardless of higher correctness
Code-only quality threshold	Fictional 80% threshold for the local gate model, not a universal release rule
Gate model	Explicit Boolean quality, privacy, compatibility and recovery results; all required
Limits	A compatibility check or all supplied gates passing supports review only; no real migration, adapter, deployment or external-effect undo

Name supplied and missing evidence

> **List the given facts and one fact that is not supplied.**
>
> _____
>
> _____
>
> _____
>
> _____

▼ **Earlier wording, alternatives and picture transcripts**

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

MAIN LESSON · SECTION 3

Trace the proposed old reader against the current inventory

> Rollback does not automatically rewind the data.

Original v2-write/v1-read mismatch

Version v2 writes schema 2. Version v1 reads only schema 1. A **rollback** to v1 after v2 writes data cannot safely proceed without migration or a compatible reader.

Complete reader-by-schema compatibility matrix

Main current data schema	v1 reader supports it?	v2 reader supports it?
1	Yes	Yes
2	No	Yes
2	No	Yes
Proposed code-only v1 rollback	Blocked: schema 2 remains stored	No data conversion performed
Retain v2 reader	Passes only this declared schema check	Not a full recovery approval

Main current data schema	v1 reader supports it?	v2 reader supports it?
Possible compatible recovery	Needs a tested compatible reader or approved migration/adapter	No such operation is executed here

State dependencies before judging

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Explain the visual

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ **Earlier wording, alternatives and picture transcripts**

Release and rollback — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
After deployment v2 creates schema 2 records; operator proposes rollback code alone to v1.	Version v2 writes schema 2. Version v1 reads only schema 1. A rollback to v1 after v2 writes data cannot safely proceed without migration or a compatible reader.	Code-only rollback is incompatible with stored schema 2. Use the tested recovery plan or compatible migration; preserve data and failure evidence.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Connect the picture and the explanation

Predict what a fresh privacy pass does and does not settle

> Other mandatory gates remain required.

Current task and response

- After deployment v2 creates schema 2 records; operator proposes **rollback** code alone to v1. Show the intermediate steps and give the conclusion with its scope.

> **Keep the procedure fixed. Explain what changed in the inputs and result.**

Predict the changed privacy evidence

- Change: Repair B and pass privacy on a fresh reserved case.

▼ **Check after predicting**

Prediction: B may enter the remaining gates; one repair does not approve everything.

Reason: All required release conditions still apply.

▼ **Earlier wording, alternatives and picture transcripts**

Compare a familiar case with a changed case

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Compare a higher score with a mandatory privacy gate

> Keep rollback compatibility separate from the pictured quality comparison.

Main code-only rollback failure and checked conclusion

After deployment v2 creates schema 2 records; operator proposes **rollback** code alone to v1.

Code-only **rollback** is incompatible with stored schema 2. Use the tested recovery plan or compatible migration; preserve data and failure **evidence**.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Separate A/B scores and complete gate qualification

A higher score can still block release. Version B is proposed to replace A; every privacy gate is mandatory.

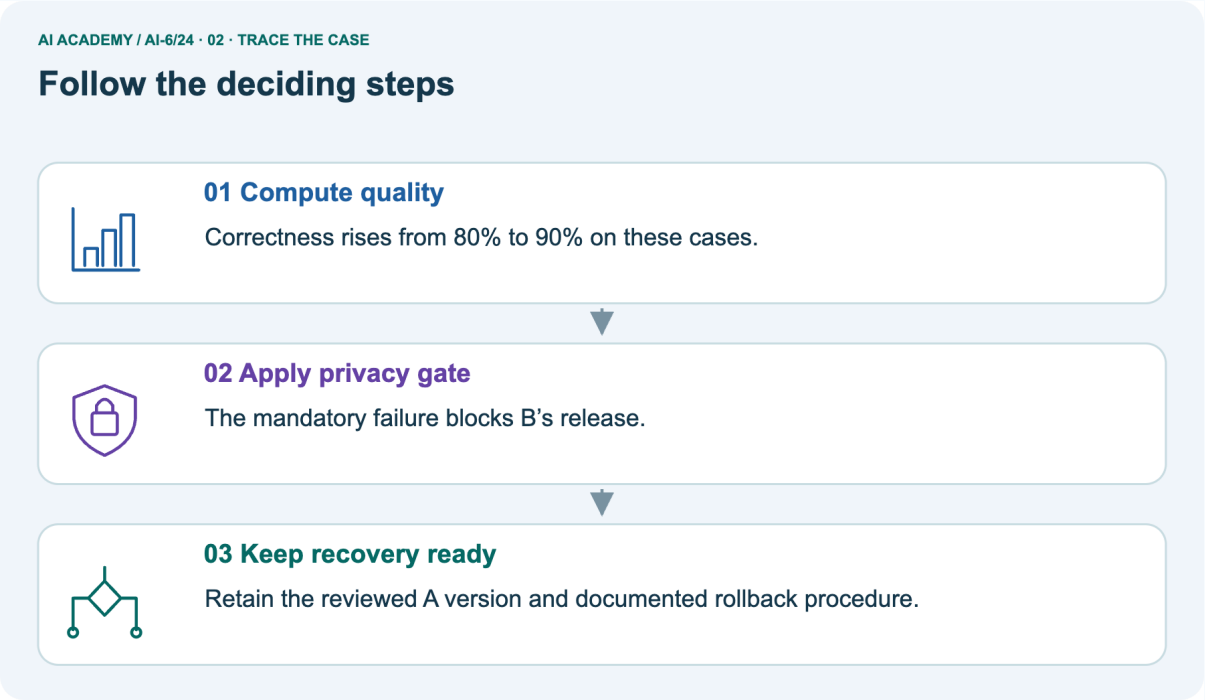
A passes 8/10 correctness cases. B passes 9/10 but fails one mandatory privacy check.

The main rollback example concerns reader compatibility with schema 2. The separate picture compares A and B on ten correctness cases: 8/10 is 80%, 9/10 is 90%, a rise of 10 percentage points. B’s mandatory privacy failure still blocks release.

Repairing B and passing a fresh reserved privacy case adds evidence for that privacy check; it does not prove that every other required check passed or that all privacy risks are resolved. Keep the original failure and the repaired version’s new evidence.

Retaining reviewed A and a documented recovery plan is useful only if its dependencies remain compatible with current stored data; neither rollback nor this picture reverses external effects.

Trace score, mandatory gate and compatible recovery



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Compute quality	Correctness rises from 80% to 90% on these cases.
2. Apply privacy gate	The mandatory failure blocks B's release.
3. Keep recovery ready	Retain the reviewed A version and documented rollback procedure.

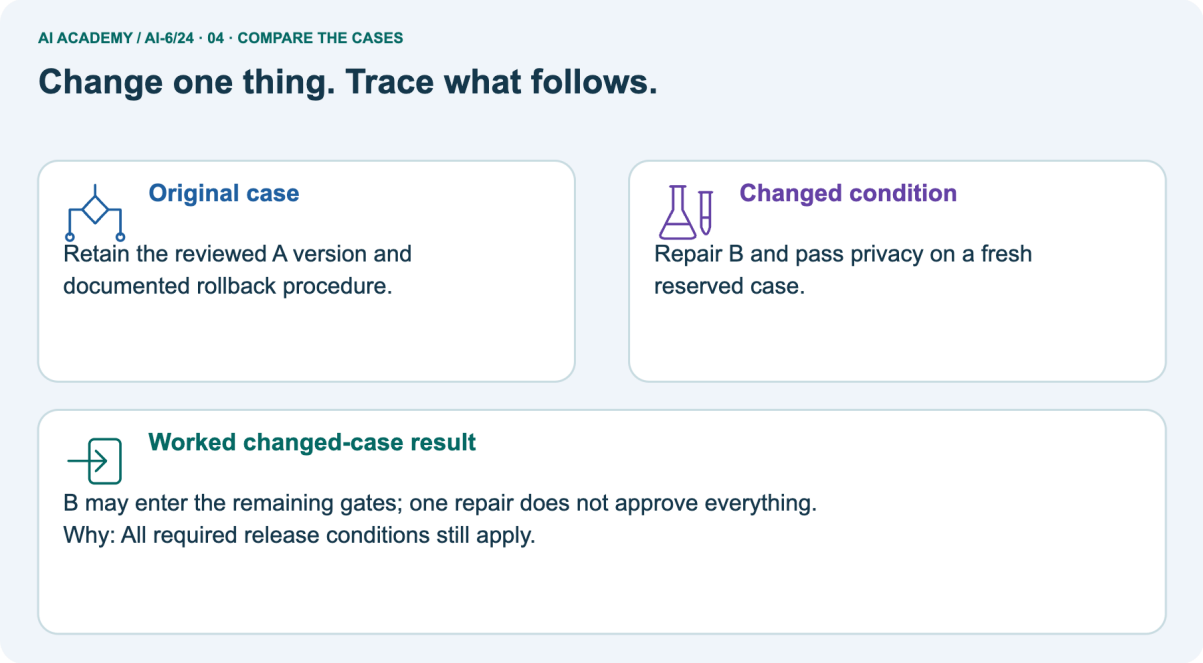
Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Release criteria combine benefits and mandatory conditions. **Rollback** requires known versions and compatible data, not an informal promise.

Predict a fresh privacy pass

- Change one thing: Repair B and pass privacy on a fresh reserved case. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain remaining gates and external-effect limits

- Rolling back code may not reverse earlier external effects or incompatible data changes.

- Read the labelled picture: Compute A's and B's correctness percentages. Why is B still blocked, and does passing a fresh privacy check automatically satisfy every release gate?

▼ Earlier wording, alternatives and picture transcripts

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Release and rollback

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Audit compatibility and mandatory conditions without averaging them away

> Preserve version dependencies and original failures.

Core release qualification and failure analysis

A **rollback** plan needs testing; an available old binary is not sufficient recovery **evidence**.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Complete dependency and gate probes

Probe	Expected interpretation
Old code cannot read new stored schema	Block code-only rollback
Reader supports all observed schemas	Passes that compatibility check only
Higher correctness but required privacy failure	Release blocked
Privacy repair passes one fresh case	New scoped evidence; other mandatory conditions still required
Missing gate or non-Boolean gate value	Reject incomplete decision record
All declared fixture gates pass	Eligible for review, not automatic deployment or universal assurance
Schema migration or adapter proposed	Must be separately tested and authorized; not assumed available
External action already completed	Code rollback does not automatically undo it
Old prompt needs fields absent from new corpus	Dependency mismatch; restore compatible combination or tested adapter

Guided repair and explanation

- A report omits this qualification: A **rollback** plan needs testing; an available old binary is not sufficient recovery **evidence**. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

State one repair and the checks you would rerun.

▼ **Earlier wording, alternatives and picture transcripts**

Locate the earliest unsupported step

MAIN LESSON · SECTION 8

Practise evidence-based release decisions

> Name the blocking condition rather than relying on the best-looking score.

Current explanation and guided check

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Explain your prediction to a partner before comparing results.

Supported W2 trace

Use workbook W2 for the connected example “Release and **rollback**”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Individual scope check

- Use the worked case for Release and **rollback**. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Picture response

- Compute A’s and B’s correctness percentages. Why is B still blocked, and does passing a fresh privacy check automatically satisfy every release gate?

My explanation using a pictured step and the given facts

Reserve the selected prompt/corpus case

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ Earlier wording, alternatives and picture transcripts

Practise with less help: Trace the worked case

Check your reasoning

Explain the picture

MAIN LESSON · SECTION 9

Run a complete compatibility and gate-decision model

> Check scoped decisions without turning them into release actions.

Current code-only rollback investigation

After deployment v2 creates schema 2 records; operator proposes **rollback** code alone to v1. Show the intermediate steps and give the conclusion with its scope.

Runnable local compatibility and gate model

```
"""Pure fictional compatibility and mandatory-gate decisions; no release operations."""
from copy import deepcopy
READERS={'v1':{1},'v2':{1,2}}
RELEASES={'A':{'code':'v1','config':'cfg1','schema':1,'prompt':'p1','corpus':'c1','evidence':'eval-A'},
          'B':{'code':'v2','config':'cfg2','schema':2,'prompt':'p2','corpus':'c2','evidence':'eval-B'}}
def rollback_check(target_code,stored_schemas):
 if target_code not in READERS:raise ValueError('unknown target reader')
 if type(stored_schemas) is not list or not stored_schemas:
  raise ValueError('nonempty observed schema inventory required')
 if any(type(s) is not int or s not in (1,2) for s in stored_schemas):raise ValueError('unknown schema')
 incompatible=sorted(set(stored_schemas)-READERS[target_code])
 return {'status':'blocked' if incompatible else 'compatible_on_declared_schema_check',
         'incompatible_schemas':incompatible,'target_code':target_code}
GATES={'quality','privacy','compatibility','recovery'}
def release_decision(gates):
 if type(gates) is not dict or set(gates)!=GATES or any(type(v) is not bool for v in gates.values()):
  raise ValueError('complete explicit Boolean gate results required')
 failed=sorted(k for k,v in gates.items() if not v)
 return {'eligible_for_review':not failed,'failed_gates':failed}
def correctness(passed,total):
```

```
 if type(passed) is not int or type(total) is not int or total<=0 or not 0<=passed<=total:
  raise ValueError('valid integer pass count and positive denominator required')
 return 100*passed/total

stored=[1,2,2];before=deepcopy(stored)
main=rollback_check('v1',stored)
compatible_reader=rollback_check('v2',stored)
assert main['status']=='blocked' and main['incompatible_schemas']==[2]
assert compatible_reader['status']=='compatible_on_declared_schema_check' and stored==before
# Separate picture supplies correctness counts and a mandatory privacy failure.
quality_a=correctness(8,10);quality_b=correctness(9,10)
# This exercise declares a fictional quality threshold of 80 percent, not a universal rule.
initial_gates={'quality':quality_b>=80,'privacy':False,'compatibility':True,'recovery':True}
initial=release_decision(initial_gates)
assert quality_a==80 and quality_b==90 and initial['failed_gates']==['privacy']
# Fresh privacy evidence is only one gate: other gates may be incomplete or fail.
privacy_repaired_only=release_decision({'quality':True,'privacy':True,'compatibility':False,'recovery':True})
all_declared_pass=release_decision({key:True for key in GATES})
assert not privacy_repaired_only['eligible_for_review'] and all_declared_pass['eligible_for_review']
print('Main code-only rollback:',main,'compatible reader check:',compatible_reader)
print('Picture A/B correctness:',quality_a,quality_b,'B initial decision:',initial)
print('Privacy repaired but compatibility fails:',privacy_repaired_only)
print('All declared gates pass:',all_declared_pass,'not an automatic release or complete real-world
assurance')
print('Release manifests:',RELEASES)
print('No migration, restoration, adapter, external-effect reversal or deployment is executed; source
inventory unchanged:',stored==before)
```

Expected outcomes and limits

1. Predict v1's decision against current schemas [1, 2, 2], then compare the v2 reader check. The source inventory remains unchanged. Compute A/B correctness before inspecting B's privacy failure. The explicit fictional gate model first blocks B on privacy, then shows a separate state where privacy passes but compatibility fails, then shows all supplied gates passing as eligibility for review only.

   The latter states are constructed variants, not measurements of a repaired live system. No migration, restoration, adapter or release runs. Use "not run" for a paper trace and preserve the original failed decision separately.

Evidence ledger and reflection

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or "not run"	

Record	Your evidence
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ **Earlier wording, alternatives and picture transcripts**

Predict, investigate and preserve the evidence

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: old prompt against a changed corpus

> Check required fields before proposing a one-component revert.

Complete selected independent task

New release changes corpus and prompt. Old prompt requires old field names absent from the new corpus. Is reverting prompt alone valid?

First attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Apply a new case independently

Attempt this without hints

New release changes corpus and prompt. Old prompt requires old field names absent from the new corpus. Is reverting prompt alone valid?

> **My independent procedure, result and limitation**
>
> | |
> |---|
> | |
> | |
> | |
> | |

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

MAIN LESSON · SECTION 11

Save one manifest with all release dependencies

> Keep old failure evidence and new repair evidence versioned.

Project task and artifact

Add a labelled release and **rollback evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of release and **rollback**.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Actual run or trace and limitations

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

▼ **Earlier wording, alternatives and picture transcripts**

Evidence for your project

MAIN LESSON · SECTION 16

Explain the compatible combination and the remaining gates

> A recovery claim must identify which versions and data it covers.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of release and **rollback**

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Use service vocabulary precisely**

REFERENCE · SECTION 5

Reference · Use service vocabulary precisely

> Connect each term to an input, operation, evidence source or outcome.

Vocabulary and your explanation

Word	Meaning
Release manifest	Versions and artifacts included in a release
Compatibility	Whether components can correctly use one another's contracts
Rollback	Returning to an earlier compatible operating version

Use two terms to explain a step in this week's example.

▼ **Extra challenge · Explain a stronger case**

REFERENCE · SECTION 10

Extra challenge · Explain a stronger case

> Choose the next challenge after the main route.

Explain,repair and extend

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.
- A report omits this qualification: A **rollback** plan needs testing; an available old binary is not sufficient recovery **evidence**. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.
- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

Explain → apply → challenge

> **Explain which evidence supports your conclusion and which claim remains unproved.**

▼ **Reference · Check the service reasoning and its limits**

REFERENCE · SECTION 12

Reference · Check the service reasoning and its limits

> Use after practice; keep the selected independent case for its first attempt.

Current explanation and response

- Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

  For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

> **Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.**

> **My limited claim and a useful next check**

▼ **Extra practice · Revisit the complete trace**

REFERENCE · SECTION 13

Extra practice · Revisit the complete trace

> Predict before reading the supplied steps.

Complete trace and evidence

Complete evidence record

Release and rollback — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
After deployment v2 creates schema 2 records; operator proposes rollback code alone to v1.	Version v2 writes schema 2. Version v1 reads only schema 1. A rollback to v1 after v2 writes data cannot safely proceed without migration or a compatible reader.	Code-only rollback is incompatible with stored schema 2. Use the tested recovery plan or compatible migration; preserve data and failure evidence.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

> **My complete trace of the worked case**

▼ **Reference · The earlier route checklist**

REFERENCE · SECTION 15

Reference · The earlier route checklist

> Use one main route rather than repeating the lesson.

Targets,route and first prediction

Start with this question: A higher score can still block release. Version B is proposed to replace A; every privacy gate is mandatory.

By the end, I can	How I will show it
Explain the mechanism and terminology of release and rollback	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ **Extra practice · Predict,check and repair**

REFERENCE · SECTION 17

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- A higher score can still block release. Version B is proposed to replace A; every privacy gate is mandatory.
- Which labelled step matches this detail? Retain the reviewed A version and documented **rollback** procedure.
- 1. Keep recovery ready
- 2. Apply privacy gate

- 3. Compute quality
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Keep recovery ready Retain the reviewed A version and documented **rollback** procedure. Release criteria combine benefits and mandatory conditions.

Rollback requires known versions and compatible data, not an informal promise.

Changed-case question and optional hint

- Changed-case practice: Repair B and pass privacy on a fresh reserved case. Which conclusion is supported by the supplied facts?
- 1. B may enter the remaining gates; one repair does not approve everything.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ Check after the changed attempt

After your attempt, compare with the supported explanation: B may enter the remaining gates; one repair does not approve everything. All required release conditions still apply.

Repair and retain the record

Review Deployment environments (ai-6/23). Goal: Explain the **mechanism** and terminology of deployment environments. Short practice: Identify the staging data and credential boundary.

Why must the rehearsal stop if its credential reaches production even though the **process** is labelled staging?. Return to this week and explain what you changed.

Review Caching and stale information (ai-6/21). Goal: Explain the **mechanism** and terminology of caching and stale information. Short practice: Identify the version mismatch in the cached 09:00 answer.

What should the first version -5 request return, and what permits a later version -5 cache hit?. Return to this week and explain what you changed.

Review Untrusted **input** boundaries (ai-6/18). Goal: Explain the **mechanism** and terminology of untrusted **input** boundaries. Short practice: Classify h3, H3, h30 and the trailing-space **input** by the stated shape contract.

Does a valid z0 grant record access, and which separate gate still remains?. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

▼ Reference · Six-picture concept journey

Reference · Six-picture concept journey

> Choose the picture that supports the current explanation.

01. Release and rollback: the case

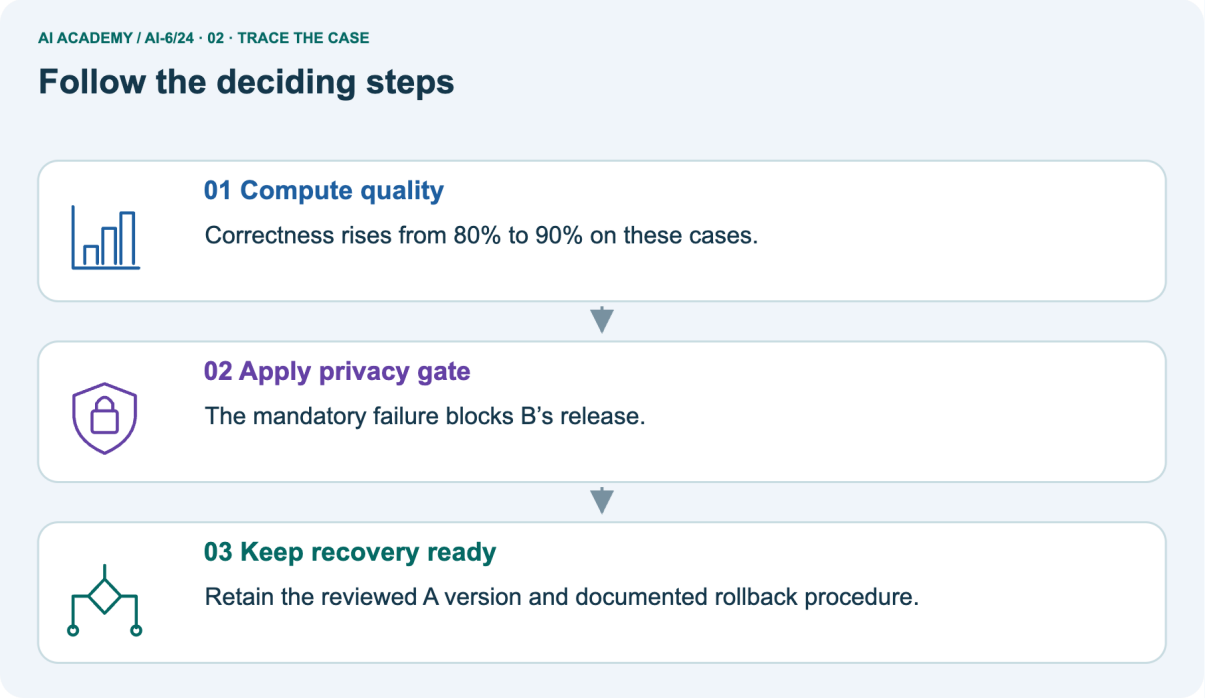


Follow the labelled picture with the matching explanation. Open the vector diagram

A higher score can still block release. Version B is proposed to replace A; every privacy gate is mandatory. Supplied facts: A passes 8/10 correctness cases. B passes 9/10 but fails one mandatory privacy check.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

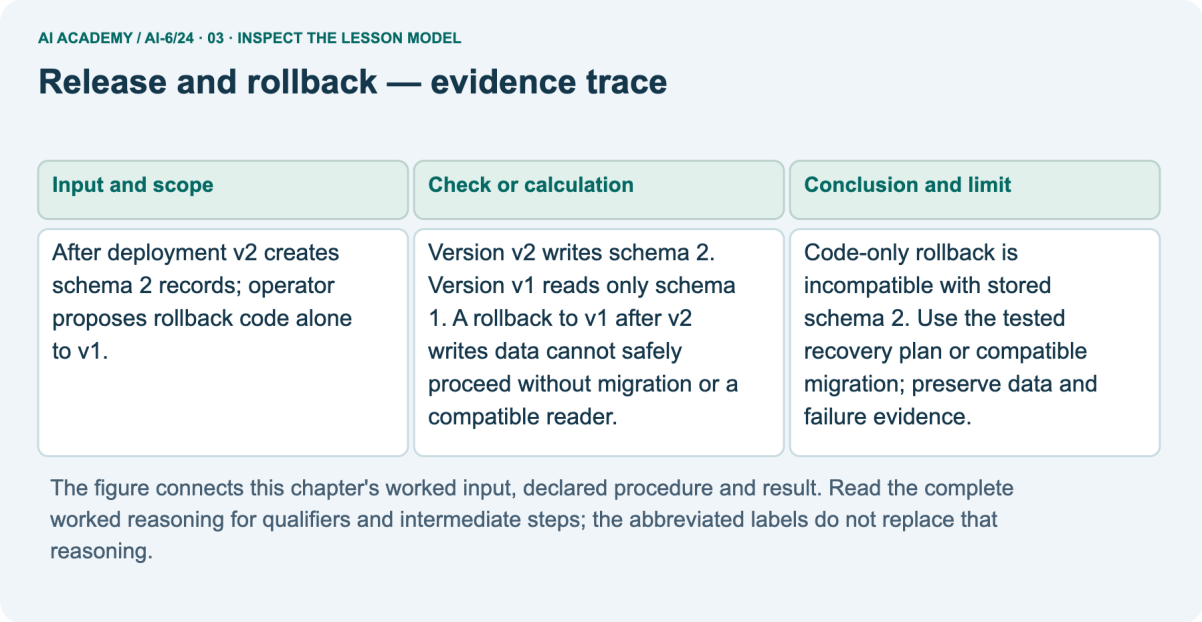


Follow the labelled picture with the matching explanation. Open the vector diagram

Compute quality: Correctness rises from **80%** to 90% on these cases. → Apply privacy gate: The mandatory failure blocks B's release. → Keep recovery ready: Retain the reviewed A version and documented **rollback** procedure.

Point to each numbered stage and explain which recorded **input** it uses.

03. Release and rollback — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

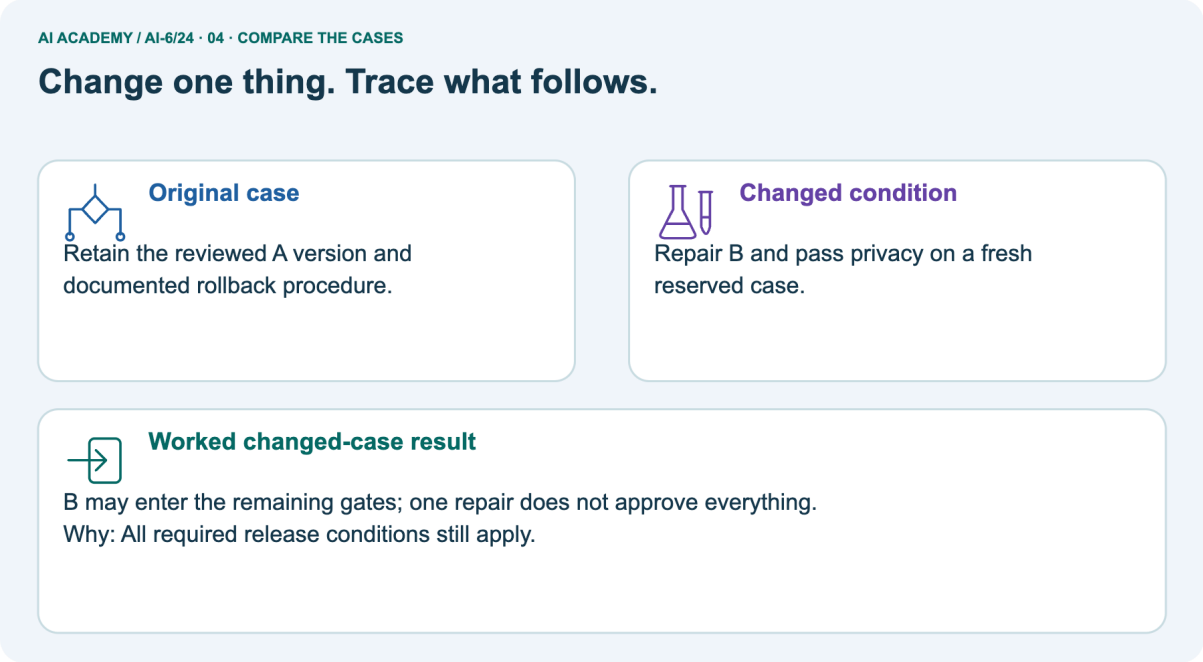
Release and **rollback** — **evidence** trace. After deployment v2 creates schema 2 records; operator proposes **rollback** code alone to v1. | Version v2 writes schema 2. Version v1 reads only schema 1.

A **rollback** to v1 after v2 writes data cannot safely proceed without migration or a compatible reader. | Code-only **rollback** is incompatible with stored schema 2.

Use the tested recovery plan or compatible migration; preserve data and failure **evidence**.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



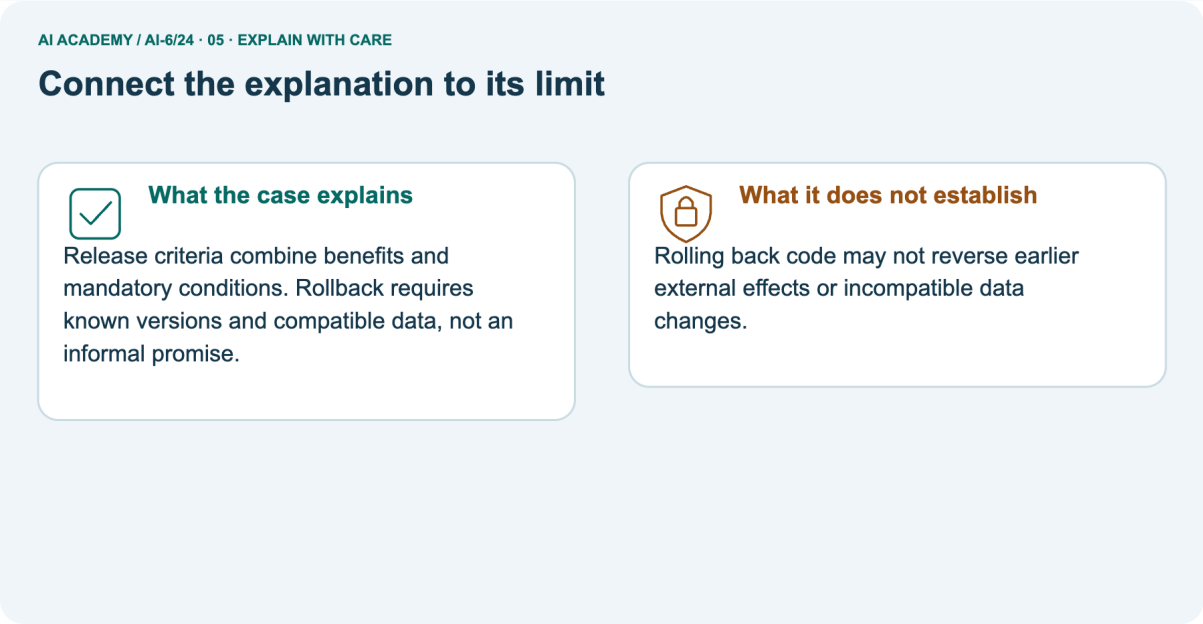
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Retain the reviewed A version and documented **rollback** procedure. Changed condition: Repair B and pass privacy on a fresh reserved case.

Expected result: B may enter the remaining gates; one repair does not approve everything. Reason: All required release conditions still apply.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



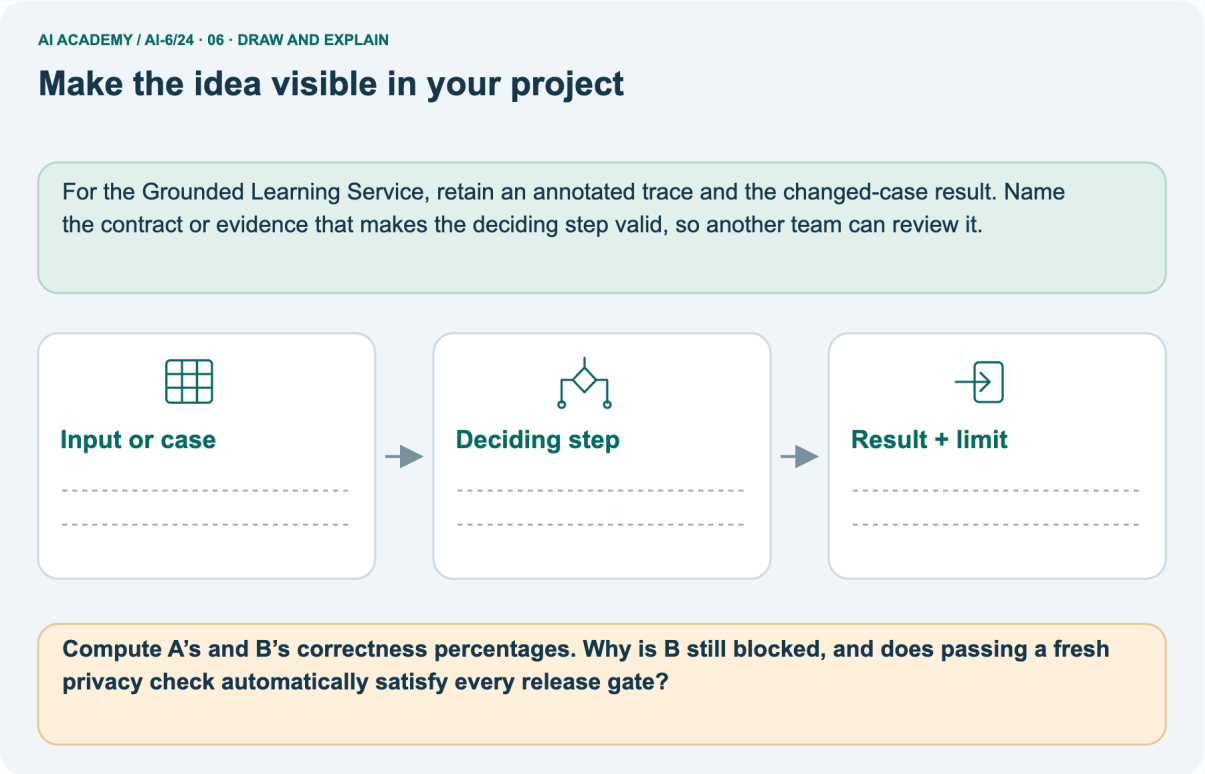
Follow the labelled picture with the matching explanation. Open the vector diagram

Release criteria combine benefits and mandatory conditions. **Rollback** requires known versions and compatible data, not an informal promise.

Limit: Rolling back code may not reverse earlier external effects or incompatible data changes.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Compute A's and B's correctness percentages. Why is B still blocked, and does passing a fresh privacy check automatically satisfy every release gate?

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Release and rollback: the case

02. Follow the deciding steps

03. Release and rollback — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.